# PSO-WORKING

**Step 1: Initialization**

Randomly initialize a population of particles.

Each particle has:

- **Position (Xi)** → represents a potential solution.
- **Velocity (Vi)** → direction and speed of movement.

Example:
If optimizing a function f(x, y), then
Xi = (xi, yi) and Vi = (vxi, vyi).

**Step 2: Fitness Evaluation**

Evaluate the **fitness** of each particle using the **objective function** (the function to optimize).
Example: If we are minimizing $f(x)$, fitness = $f(Xi)$.

**Step 3: Update Personal Best (pbest)**

For each particle:

Compare current fitness with the previous best fitness.
If current is better → update pbest.

## Step 4: Update Global Best (gbest)

Among all particles, choose the one with the **best fitness** so far → set as **gbest**.

## Step 5: Update Velocity and Position

This is the **core of PSO** — particles adjust their velocity based on three factors:
**Inertia (ωVi)** → tendency to continue moving in the same direction.
**Cognitive component ($c_1r_1$(pbest - Xi))** → learning from its own best experience.
**Social component ($c_2r_2$(gbest - Xi))** → learning from the group's best experience.

**Velocity update formula:**

$$V_i(t+1) = \omega V_i(t) + c_1 r_1 \big(pbest_i - X_i(t)\big) + c_2 r_2 \big(gbest - X_i(t)\big)$$

Where:

ω = inertia weight (balances exploration & exploitation)

$c_1$ = cognitive acceleration coefficient

$c_2$ = social acceleration coefficient

$r_1$, $r_2$ = random numbers in [0, 1]

**Position update formula:**

$$X_i(t+1) = X_i(t) + V_i(t+1)$$

**Step 6: Stopping Condition**

Repeat steps 2–5 until:

Maximum number of iterations reached, OR

Desired fitness (solution quality) achieved.

THANK YOU